CS 6210 Spring 2026 Test 2

(120 min Canvas Quiz)

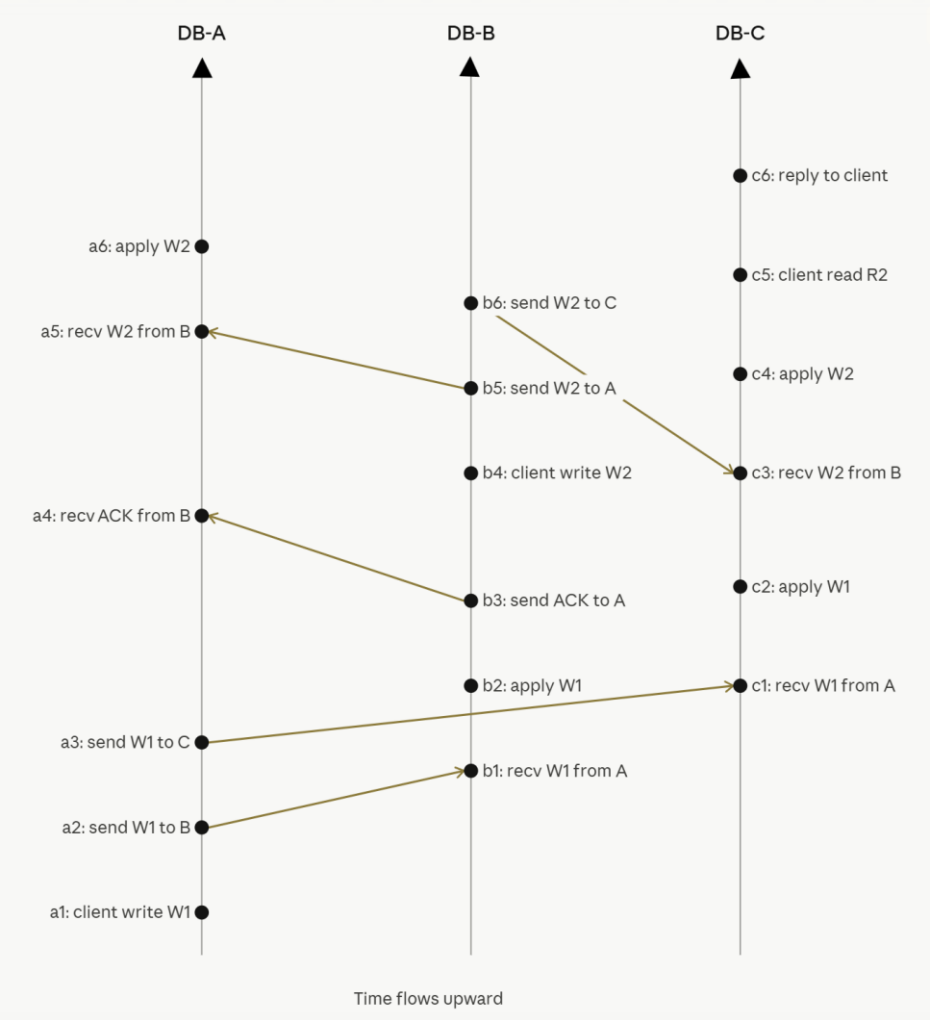
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Distributed Systems [28 points]

Lamport's Logical Clock [7 pts]

1. [7 points] In the questions below, DB-A, DB-B, and DB-C represent a replicated database. For all questions in this section, explain your reasoning in one or two sentences. Correct justification is required to receive full credit.



a) [2 points] At event c5, which writes are guaranteed to already be reflected at DB-C: W1, W2, both, or neither? Explain.

b) [2 points] At event a4, can DB-A conclude that all three database nodes have already seen W1? Explain.

c) [3 points] Assume the replicated database stores a single key x.

- Write W1 updates x = 1
- Write W2 updates x = 2

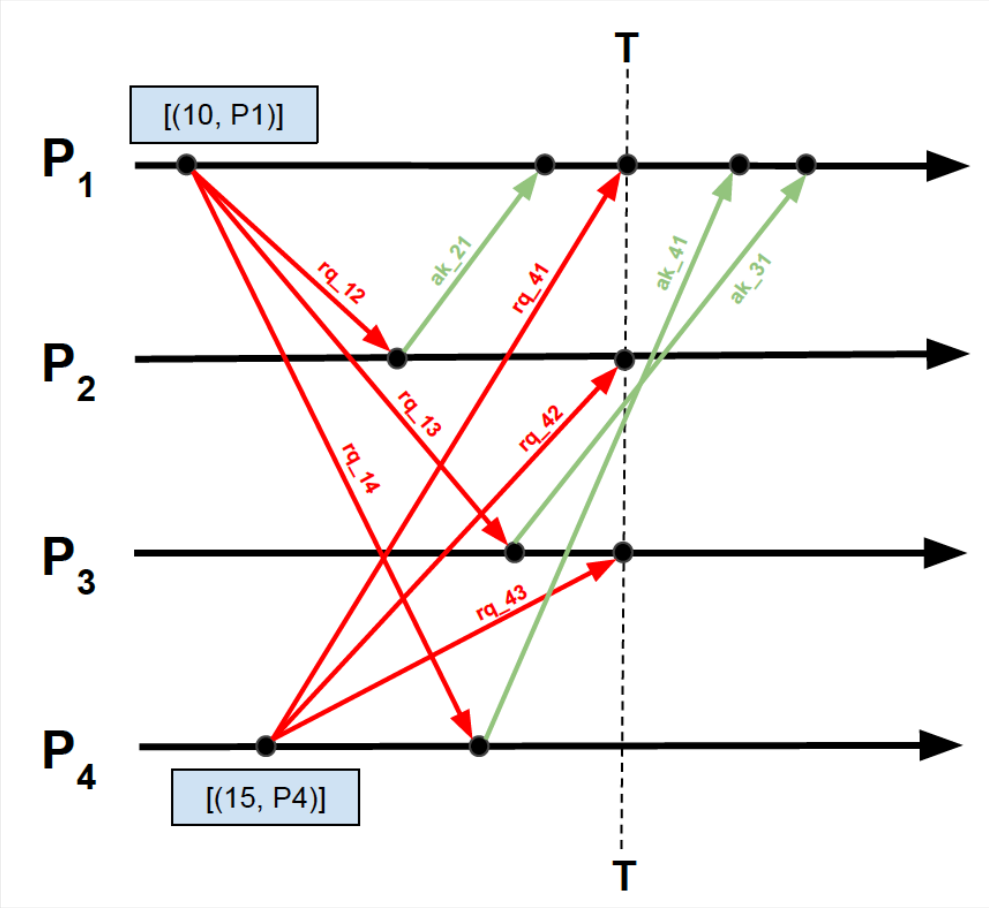
A read is called globally consistent if, at the time it is served, all replicas would return the same value of x.

Now suppose we add a new event Rc, on DB-C, where Rc is a client read of x, and it occurs after c3 but before c4.

Will the read at Rc be globally consistent? State yes or no and explain your reasoning in one or two sentences.

Lamport's ME Algorithm [8 pts]

1. [8 points] Consider a distributed system with 4 process nodes (P1, P2, P3 and P4) implementing a distributed Lamport's mutual exclusion algorithm. The figure illustrates the timeline for process nodes requesting locks and when they receive different types of messages (request, acknowledgement, and release).

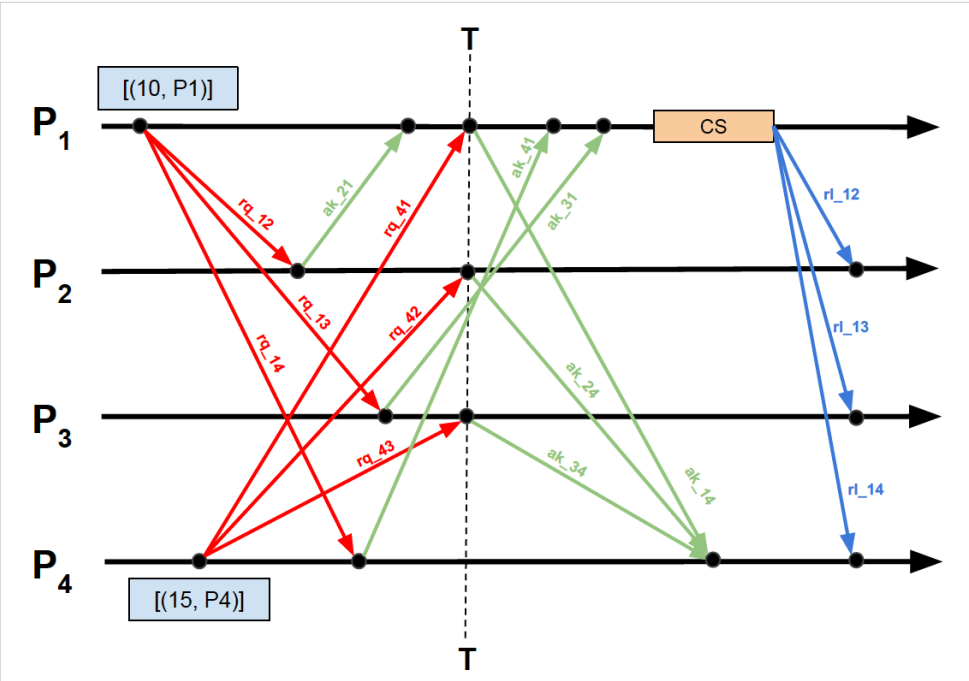


In the above figure,

- Red arrows (->) represent a lock request message
- Green arrows (->) represent an acknowledgment message
- Blue arrows (->) represent a lock release message
- The initial state of queues for processes P1 and P4 have been given in the light blue box. Each element in the queue holds the timestamp and the process number.
- Messages are represented using the following convention:
    - Lock request denoted by *rq*, followed by the node number from which the message was sent and finally the node number which received the message.
    - In a similar manner, acknowledgment messages are denoted by *ak* + the above logic.
    - Lock release messages are denoted by *rl* + the above logic.
- An orange box denotes the critical section after the process has acquired the lock.

a) [2 points] Can P1 acquire the lock at the global timestamp T (denoted by vertical dotted line)? If yes, justify your answer based on the two conditions required to enter the

critical section. If no, please provide justification based on the same condition. Also, if your answer is no, what could possibly go wrong if P1 indeed acquired the lock? No credits for simply stating yes/no.



b) [2 points] The standard Lamport's ME algorithm requires 3(N-1) messages per lock acquisition in a system of N nodes (N-1 messages each for lock request, acknowledgement, and release).

(i) Can you suggest an improvement which helps to reduce communication messages without violating safety?

(ii) If an improvement is possible, give ONE example of a message that can be discarded in our current timeline diagram with justification. (No points without justification)

c) [2 points] Your colleague suggests another improvement – we send lock release messages ONLY to nodes that are currently awaiting to use the lock. Does this improvement honor safety and correctness? Provide your justification. What messages could be discarded in our timeline if we were to implement the improvement?

d) [2 points] Analyze the message complexity of the algorithm after the last improvement in the following scenarios:

- Low contention: Only one node ever requests a lock at any given time.
- High contention: All N nodes actively request to acquire locks.

How many messages are required in each case in terms of N? No credit without justification.

RPC Latency Limits [6 pts]

1. [6 points] Some client process C is making an RPC call to server process S. The amount of time for this RPC call to execute, t, can be represented as an expression of the following variables:

t_c – the time to copy *n* bytes

t_x – the time to context switch between process *P* and *P'*

t_n – the time to transmit *n* bytes over the network

t_s – the time for procedure P to execute on S

You may assume these are the only components of latency and that the client's arguments and server's response are both *n* bytes.

a) [2 points] Assuming no optimizations, provide an expression for t. Explain your reasoning.

b) [2 points] Additionally, the kernel guarantees that any context switches occur after any data is transmitted. Provide an updated expression for t. Explain your reasoning.

c) [2 points] Additionally, the kernel provides a buffer to be directly used by the client and/or server stubs. With this additional optimization, provide an updated expression for t. Explain your reasoning.

Active Networks [7 pts]

1. [4 points] Consider three active nodes A → B → C in sequence. Capsule originates at A and passes through B before reaching C. Node B currently has the code for this TYPE in its soft store, but node C has not seen this TYPE before. A new capsule of this TYPE now arrives at C.

a) [2 points] Walk through exactly what happens at C, step by step, until the capsule is either processed or dropped.

b) [2 points] Now suppose that, unlike in (a), B's soft store was recently cleared due to a router restart and no longer has the code for this TYPE. How does this change the outcome at C? What does this tell us about the robustness of the PREV-based design?

2. [3 points] A student proposes embedding full code directly in every capsule instead of using a TYPE field and soft store. Give one advantage and one disadvantage of this proposal.

Distributed Objects and Middleware [10 points]

Spring OS [7 pts]

1. [7 points]

You are a software engineer on a team that maintains an important application that runs on Spring OS. Currently, Client Domain A, Client Domain B, Server Domain Y and Server Domain Z all run on a single node. Assume Server Z exports Object Z and Server Y exports Object Y.

Recently, the entire application has been running slow. To improve system performance and scalability, your manager has proposed partitioning the domains across two nodes connected by a LAN.

Proposed Split:

- Node 1: Client Domain A, Server Domain Z
- Node 2: Client Domain B, Server Domain Y

a) [2 points] Your coworker claims that the client-side stubs must be changed for accessing the servers on remote nodes (e.g., Client A accessing Server Y". What is the fallacy in your coworker's claim? What does change on the client side for this proposed split to work?

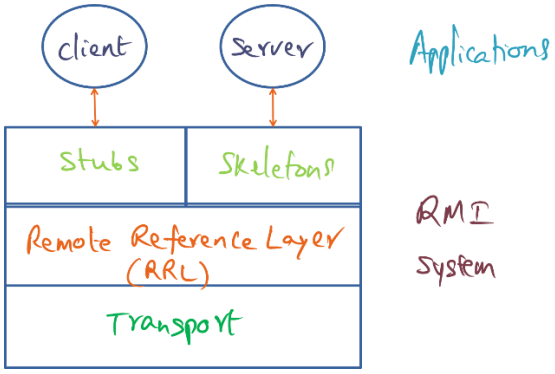
b) [3 Points] Object Y supports both read and write operations. Client A must have read/write access to Y, while Client B must have read only access to Y. Using succinct bullets, state how the Spring mechanisms (capabilities, access control, doors) allow achieving the desired objectives.

c) [2 Points] Suppose Client A later passes its read/write capability for Object Y to Client B. With a succinct bullet, explain how Client B gains write access even though the original privilege level for Client B was only read access.

Java RMI [3 pts]

1. [3 points]

A server exports a remote object. A client connects, performs several operations, and then the network connection between the client and server is abruptly severed (a network failure). The client application has implemented semantics to attempt to reconnect to the server. Assume the client's reconnection is successful. No error is propagated to the client application.



With reference to the figure above, explain with concise bullets the role of the different layers of the RMI implementation stack for client to handle this network failure and reconnecting to the server.

Distributed Subsystems [32 points]

Global Memory Systems [11 pts]

1. [5 points]

Based on the Global Memory Systems (GMS) architecture, analyze the following multi-node scenario. Use the provided terminology for your answers. For any given page:

- PFD States: **Local-Private** – node owns an exclusive local copy, **Local-Shared** – multiple nodes hold shared copy, **Global-Private** - node holds the page in its global memory on behalf of another node, **Non-existent** – node does not hold any copy.
- GCD: **Home Node** for a given page; given a UID it returns the node IDs that are currently hosting the page in set notation, e.g., {X, Y}

Initial Scenario (Time T0)

A 3-node cluster consists of Node A, Node B, and Node C. For the data page P, Node A is the Home Node. At T0, Node A has just loaded Page from disk into its local part of the DRAM for a process. Nodes B and C are currently idle.

Current Status at T0 with respect to page P:

1. GCD at Home Node for P: {A}
2. PFD at Node A: Local-Private
3. PFD at Node B: non-existent
4. PFD at Node C: non-existent

a) [2 points] Event 1 (Time T1>T0)

A process on Node B page faults on Page P.

Provide the status for the following 4 entries after Event 1:

1. [0.5 points] GCD at Home Node for P: (List) _____
2. [0.5 points] PFD at Node A: (State) _____
3. [0.5 points] PFD at Node B: (State) _____
4. [0.5 points] PFD at Node C: (State) _____

b) [1 point] Event 2 (Time T2>T1)

Node A experiences memory pressure and intends to evict Page P. The page is clean (i.e., not dirty). Node C would be the candidate node to send the page. Should Node A decide to keep the page in the cluster.

(Answer True/False with justification. No credit without justification):

Node A will send the Page P to Node C’s global part of the DRAM.

c) [2 points]

Event 2 (Time T2>T1)

Node A experiences memory pressure and intends to evict Page P. The page is clean (i.e., not dirty). Node C would be the candidate node to send the page. Should Node A decide to keep the page in the cluster.

Provide the status for the following 4 entries after Event 2:

1. [0.5 points] GCD Owner List at Home Node: (List) _____
2. [0.5 points] PFD at Node A: (State) _____

3. [0.5 points] PFD at Node B: (State) _____

4. [0.5 points] PFD at Node C: (State) _____

2. [2 points] (Answer True/False with justification; no credit without justification)

- In GMS, when a node decides to send a page to a peer node's global part of the DRAM, it writes the page to the disk if the page is dirty

3. [2 points] (Answer True/False with Justification; No credit without justification}

A simpler and more efficient design in GMS would be to broadcast the UID of the missing page to all the peers upon a page fault at a node.

4. [2 points]

Your co-worker argues that a much simpler and more efficient design in GMS would be to have a replicated data structure at each node:

- UID -> Node that is currently hosting the page

Her argument is that in one network hop, a node can get to the peer node where the missing page is currently hosted.

Give two succinct bullets, to counter her argument.

DSM [13 pts]

1. [5 points] Answer the following about the TreadMarks DSM system.

a) [5 points] In a TreadMarks DSM system, assume page X is replicated on all 8 nodes. Lock L1 is acquired in order by N1, N2, ..., N8. N1 writes X under L1. Nodes N2 through N7 acquire L1 but do not access X. N8 later acquires L1 and reads X. Which performs better in this scenario - Eager Release Consistency or Lazy Release Consistency? Justify your answer. Also, calculate the total number of DSM-specific

messages(invalidations, updates or diff requests/responses) in both ERC and LRC.

2. [8 points] Consider the following sequence of actions in a TreadMarks DSM system. Assume pristine copies of pages X and Y are at their owner node O throughout. Assume X and Y are initially not present on any other node.

T1: Processor P1:

acq(L1)
write X
rel(L1)

T2: Processor P2:

acq(L1)
write Y
rel(L1)

T3: Processor P3:

acq(L2)
write X
rel(L2)

T4: Processor P4:

acq(L1)
read X
write Y
rel(L1)

a) [2 points] What actions occur when P4 acquires lock L1?

b) [4 points] While executing the critical section under L1, P4 first reads X and then writes Y. What actions are needed to ensure correct execution?

c) [2 points] What happens when P4 releases L1?

Distributed File Systems [8 pts]

1. [2 points] xFS assigns a newly created file to a manager on the same machine as the creating client ("First Writer" policy). Simulation results in the paper show that this policy significantly reduces total network hops compared to centralized management, and that most of the improvement comes from writes rather than reads. Explain why First Writer helps writes much more than reads with succinct bullets.

2. [3 points] You are extending xFS to run on a much larger cluster with hundreds of storage servers.

A teammate proposes simplifying the design by removing stripe groups and instead striping each log segment across all storage servers in the system. Assume each client still generates the same volume of writes as before.

You evaluate this proposal and identify potential performance and reliability issues. With concise bullets, give any three distinct issues this design would cause.

3. [3 points] xFS caches index nodes at managers but explicitly does not cache them at clients.

   a) [1 point] If index nodes were cached at clients, what performance benefit would clients gain? Be specific about which network messages would be eliminated.

   b) [2 points] Despite this benefit, xFS rejects client-side index node caching. Give two distinct drawbacks of client-side index node caching from the paper.

HIDDEN - Potpourri [15 pt]

1. [2 points]

(Answer True/False with justification. No credit without justification)

In Lamport's mutual exclusion lock algorithm, when a node "A" determines that it has the lock, the contents of the local queues on all the nodes are identical.

2. [2 points] [True/False] Doubling the soft store size at an active router is unnecessary because any evicted breadcrumb can always be recovered by requesting the code from the PREV node.

3. [2 points] [True/False; No Points without Justification]

Two users, Alice and Bob, interact with a remote server on the Internet from their respective personal machines. The distributed system comprises servers and individual actors using Lamport's Logical clock for ordering requests breaking ties (if any) with the machine ID of the requests. Alice submits request A. After submitting her request, she calls Bob on the phone and asks him to submit request B from his machine. Alice's expectation is that her request will be satisfied before Bob's.

Assume:

- no message loss
- messages between any pair of machines are delivered in order.

Will the remote server fulfill Alice's expectation? State yes or no and explain your reasoning.

4. [4 points] Based on the EJB paper state one advantage (Pro) and one disadvantage (Con) each for the two EJB architectural features, namely Entity Beans and Session Façade

5. [5 points] Answer the following about the TreadMarks DSM system.

a) [2 points] In TreadMarks' multiple-writer protocol, what actions occur when a processor performs the first write to a shared page in an interval?g

b) [3 points] [True or False, with justification] If two processors modify overlapping words of the same page without intervening synchronization, TreadMarks will deterministically merge both updates by applying the diffs in order.